

$$\int (xy + y^2) dx$$

$$\int (xy + y^2) dx = \int xy dx + \int y^2 dx$$

$$= y \int x dx + y^2 \int 1 dx$$

$$= y \left[\frac{x^2}{2} \right]_0^1 + y^2 \left[x \right]_0^1$$

$$= y \left[\frac{1^2}{2} - \frac{0^2}{2} \right] + y^2 [1 - 0]$$

$$= y \left[\frac{1}{2} - 0 \right] + y^2 [1]$$

$$y \left(\frac{1}{2} \right) + y^2$$

$$\int (xy + y^2) dy$$

$$\int xy dy + \int y^2 dy$$

$$x \int y dy + \int y^2 dy$$

$$x \cdot \left[\frac{y^2}{2} \right]_0^1 + \left[\frac{y^3}{3} \right]_0^1$$

$$x \left[\frac{1^2}{2} - \frac{0^2}{2} \right] + \left[\frac{1^3}{3} - \frac{0^3}{3} \right]$$

$$x \cdot \left(\frac{1}{2} \right) + \frac{1}{3}$$

$$= \frac{x}{2} + \frac{1}{3}$$

$$\int_0^1 \int_0^2 (x+3) dy dx$$

$$= \int_0^1 \int_0^2 (x+3) \int_0^2 1 dy dx$$

$$\int_0^1 (x+3) \left[y \right]_0^2 dx$$

$$\int_0^1 (x+3) (2-0) dx$$

$$= \int_0^1 2(x+3) dx$$

$$2 \left[\int_0^1 x dx + 3 \int_0^1 1 dx \right]$$

$$2 \left[\frac{x^2}{2} \Big|_0^1 + 3 \left[x \right]_0^1 \right]$$

$$= 2 \left[\left(\frac{1}{2} - \frac{0}{2} \right) + 3(1-0) \right]$$

$$= 2 \left(\frac{1}{2} + 3 \right)$$

$$= 2 \left(\frac{1+6}{2} \right) = \frac{14}{2} = 7$$

$$\iint_R x dA$$

① express the double integral for the volume solid above $z = 8 - x - y$ and below $z = 0$, $0 \leq x \leq 8$, $0 \leq y \leq 8$

$$\Rightarrow V = \int_0^8 \int_0^8 (8 - x - y) dx dy$$

② z be the length $\frac{L \cdot W \cdot H}{L, W, H}$ rectangular box. area of base

$$\Rightarrow A = z + z + z$$

③

- ① If function f is continuous closed interval $[a, b]$, then f has absolute max value and absolute min value on $[a, b]$

Extreme value theorem

- ② function defined $y = 3 - x$ has absolute minimum value of $\underline{\hspace{2cm}}$ on interval $1 < x < 3$

→ 2

- ③ All continuous function has Max & Min value

→ True

- ④ Absolute extrema $f(x) = x^2 - 2$ is $\underline{\hspace{2cm}}$

$(0, -2)$

- ⑤ Partial derivative of $f(x, y) = 3x^2y + 2y^2$

→ $6xy$

- ⑥ number $\sqrt{3}$ is

→ Irrational

- ⑦ Equation $r^2 = a^2 \sin 2\theta$ represent

→ Lemniscate

- ⑧ Equation of $r = a \cos \theta$ represent

where a is positive

→ circle

- ⑨ $\int_0^{\pi} \int_0^1 r^2 dr d\theta$

→ $\pi/3$

(12) $\int_0^1 \int_0^1 \int_0^1 xyz \, dx \, dy \, dz = \frac{1}{8}$

(13) $r(t) = x(t)i + y(t)j$

→ vector valued function

(14) Given integral $\iint_R f(x, y) \, dx \, dy$ expressed in polar coordinate $\leq \theta < b$ and $c < r < d$

→ $\int_c^d \int_a^b f(r, \theta) r \, dr \, d\theta$

(15) Graph of equation $r = a(1 - \cos \theta)$ geometric initial line

(16) Volume of rectangular parallelepiped formula $V = xyz$

→ $dV = yz \, dz + xz \, dy + xy \, dx$

(17) A — need not be the highest point on entire graph.

→ relative Maximum

(18) Two surfaces are said to be orthogonal ^{Perpendicular}

(19) Gradient of scalar function always result —
→ vector

$$f(x, y, z) = x^3 + 2y - 3z^2 \quad \text{then } f_1, f_2, f_3 =$$

$$f_1 = 3x^2 \quad f_2 = 2 \quad f_3 = -6z$$

3. Cross product of equal and opposite non-zero vector $\Rightarrow$ zero vector

4. If $f(x, y)$ is continuous on closed and bounded
(a) absolute Max. & absolute Min

5. Geometrically magnitude of scalar triple product $\Rightarrow$ Volume of parallelepiped.

6. Gradient of scalar function defined as —
 $\text{grad } \phi = - \cancel{\nabla f(x, y)} \phi \quad \nabla \phi$

7. del operator is defined as
 $\nabla = \frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k$

8. Surface $z = f(x, y)$ falls (decreases) most rapidly at any point in the — gradient.

(26) for the function $y = f(x)$ is differentiable
 $\rightarrow \int f'(x) dx$

(27) absolute Max. is also called local Max.
 $\rightarrow$ False

(28) let m be the slope of $f(x_0, y_0)$ point is
 $\rightarrow y - y_0 = m(x - x_0)$

(29) A differentiable function of a single variable
might have _____ point
 $\rightarrow$ inflection

(30) A differentiable function of two variables
might have _____ point
 $\rightarrow$ Saddle

(31) Volume of parallelepiped with a, b and c determined
 $|\vec{a} \times \vec{b}| \cdot |\vec{c}| \cos \theta$

(32) Critical point which neither Max nor Min called
 $\rightarrow$ Saddle point

Coordinate

Every real number corresponds to — on the coordinate line.

A unique point



There is one to one correspondence between the set of point on coordinate line and set of real number

Which of following associated to each point of three dimensional spaces?

→ natural number

All axes are positive in — octant..

→ First

spherical co-ordinate of point are $(\sqrt{3}, \pi/3, \pi/2)$

where are cylindrical co-ordinates.

→ $(\sqrt{3}, \pi/3, 0)$

		<u>cylindrical</u>	<u>spherical</u>
		(r, θ, z)	(ρ, θ, ϕ)
$\rho = \sqrt{3}$	$\theta = \pi/3$	$r = \rho \sin \phi$	$r = \rho \sin \phi$
$\phi = \pi/2$	$\phi = \pi/2$	$\theta = \pi/3$	$\theta = \pi/3$
$z = \rho \cos \phi$		$z = \rho \cos \phi$	
$z = \sqrt{3} (1)$		$z = \sqrt{3} \cos \pi/2$	
$z = \sqrt{3}$		$z = 0$	
		$z = 0$	
		$(\sqrt{3}, \pi/3, 0)$	

(6) Suppose $f(x, y) = xy - 2y^2$ where, $x = 3t + 1$
and $y = 2t$ which are of following

$$\Rightarrow \frac{dt}{dt} = -4t + 2$$

$$f(x, y) = xy - 2y^2$$

$$(3t+1)(2t) - 2(2t)^2$$

$$= -2t^2 - 2t$$

$$\frac{d}{dt} = 4t + 2$$

(7) Let $w = f(x, y, z)$ and $x = g(r, s)$, $y = h(r, s)$
 $z = (r, s)$ by chain rule $\frac{\partial w}{\partial r} =$ —

$$\Rightarrow \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial r}$$

(8) Magnitude of vector $\vec{a}$ is 2, $\vec{b}$ is 3
angle between tail to tail is 45° . $\vec{a} \cdot \vec{b}$?

$$\Rightarrow 4.24$$

$$|\vec{a} \cdot \vec{b}| = |\vec{a}| |\vec{b}| \cos \theta$$

$$= 2 \cdot 3 \cdot \cos 45^\circ$$

$$= 6 \cdot (0.707)$$

$$= 4.2$$

(9) Let function $f(x, y)$ continuous at origin
 $f(x, y) = \begin{cases} 0 & \text{if } x \geq 0 \text{ and } y \geq 0 \\ 1 & \text{otherwise} \end{cases}$ ($\infty = 1/0$)

$\Rightarrow f(x, y)$ is continuous at origin.

the function $f(x, y)$ continuous at origin?
if not why?

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

$f(0, 0)$ is not defined

$$\left\{ \begin{array}{l} \frac{x^2 y}{x^2 + y^2} = \frac{0}{0} \\ \text{undefined} \end{array} \right.$$

$f(0, 0)$ is defined and $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ exist but these two number are not equal.

① Let R be a closed region in two dimensional space. what does the double integral over R calculate

→ Radius of inscribed circle of R .

② formulas can be used find the volume of parallelepiped with adjacent edges formed vector $\vec{a}, \vec{b}, \vec{c}$?

→ $|\vec{a} \cdot (\vec{b} \times \vec{c})|$ or $(\vec{a} \times \vec{b}) \cdot \vec{c}$

③ Two surface are said to be orthogonal at point if their intersection normal at point.

→ Perpendicular

(14) By extreme value theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R , then $f(x, y)$ has both

→ Absolute Maximum & absolute Min value

(15) Let function $f(x, y)$ has continuous second order partial derivative (f_{xx}, f_{yy} & f_{xy}) in some circle centered at critical point (x_0, y_0) & let $D = f_{xx}(x_0, y_0) - f_{yy}^2(x_0, y_0)$

$D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f —

→ Relative Maximum at (x_0, y_0) ✓

Point — {
 (1) Relative Maximum at (x_0, y_0) $D > 0$ $f_{xx}(x_0, y_0) < 0$
 (2) Relative Min at (x_0, y_0) $D > 0$ $f_{xx}(x_0, y_0) > 0$
 (3) Saddle point at (x_0, y_0) $D < 0$
 (4) No conclusion can be drawn $D = 0$

(16) If $R = R_1 \cup R_2$ where R_1 and R_2 are non overlapping regions then $\iint_R f(x, y) da = \iint_{R_1} f(x, y) da + \iint_{R_2} f(x, y) da$

→ $\iint_R f(x, y) da$

• $\iint_R f(x, y) da = \iint_{R_1} f(x, y) da + \iint_{R_2} f(x, y) da$

$$R = \{ (x, y) / 0 \leq x \leq 2 \text{ and } 1 \leq y \leq 4 \}$$

then $\int_R (6x^2 + 4xy^3) dA =$

$$\int_0^2 \int_1^4 (6x^2 + 4xy^3) dy dx$$

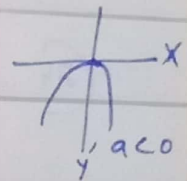
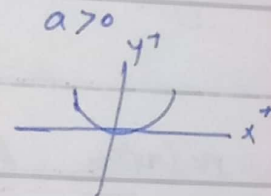
$$\Rightarrow \int_0^2 \int_1^4 (6x^2 + 4xy^3) dy dx$$

⑧ if $R = \{ (x, y) / 2 \leq x \leq 4 \text{ and } 0 \leq y \leq 1 \}$ then $\int_R (4xe^{2y}) dA =$

$$\Rightarrow \int_0^1 \int_2^4 (4xe^{2y}) dx dy$$

⑨ if $a > 0$ then Parabolas $y = ax^2 + bx + c$ open in which of following direction?

→ Positive y-direction



⑩ Rectangular Co-ordinate of point $P(1, \sqrt{3}, -2)$
What is spherical Co-ordinate?

(ρ, θ, ϕ)

$$\rho = \sqrt{x^2 + y^2 + z^2} \quad \tan \theta = \frac{y}{x} \quad \cos \phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\rho = \sqrt{1^2 + (\sqrt{3})^2 + (-2)^2} \quad \theta = \tan^{-1} \left(\frac{\sqrt{3}}{1} \right)$$

$$\rho = \sqrt{8} = 2\sqrt{2}$$

$$\theta = 60^\circ = \frac{\pi}{3}$$

$$\cos \phi = \frac{-2}{2\sqrt{2}} = -\frac{1}{\sqrt{2}}$$

$$\phi = 135^\circ$$

$$\phi = \frac{3\pi}{4}$$

$$\Rightarrow 2\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{4}$$

(21) function is not defined at some point
then it Limit exist at point

⇒ Never

(22) Suppose $f(x, y) = x^3 e^{xy}$ which one
statement is correct.

⇒ $\frac{\partial f}{\partial y} = x^4 e^{xy}$

$$\begin{aligned} \frac{d}{dy} x^3 e^{xy} \\ x^3 \cdot \frac{d}{dy} (e^{xy}) \\ x^3 \cdot e^{xy} \cdot x \\ x^4 e^{xy} \end{aligned}$$

(23) $f(x, y) = xy - y^3 + \ln x$ then $\frac{\partial^2 f}{\partial x^2}$

⇒ $2y - \frac{1}{x^2}$

$$\begin{aligned} \frac{\partial f}{\partial x} &= y - 0 + \frac{1}{x} \\ \frac{\partial^2 f}{\partial x^2} &= 2(1)y + (-\frac{1}{x^2}) \\ &= 2y - \frac{1}{x^2} \end{aligned}$$

(24) relation between the direction of gradient
at any point on surface to the tangent
plane at point

⇒ Perpendicular

(25) function $f(x, y) = \sqrt{y-x}$ is continuous in
origin and discontinuous elsewhere

⇒ $x > y$

⑥ Plane is an example of surface

⑦ Cross product of two non-zero vectors result into a vector lying in plane perpendicular both vectors

⑧ value $f(x_0, y_0)$ is an absolute maximum of $f(x, y)$ ^{domain}
 $\Rightarrow f(x_0, y_0) \geq f(x, y)$ for all $(x, y) \in D$

⑨ Dot product is not commutative, i.e. $(a, b) \cdot (c, d) \neq (c, d) \cdot (a, b)$
 $\Rightarrow$ false

⑩ absolute extrema of $f(x) = x^2 - 2$ is 0
 $\Rightarrow (0, -2)$

$$f'(x) = 2x = 0 \Rightarrow x = 0$$

⑪ Cross product of unit vector i and j ^(0, -2)
whenever the sense rotation is from i to j

$$\begin{cases} i \times j = k \\ j \times i = -k \end{cases}$$

$\Rightarrow -k$

⑫ Geometrically the magnitude of scalar Triple product gives the volume of parallelepiped

(33)

All continuous functions have Min and Max values.

→ True

(34)

For function $z = f(x, y)$ the total differential dz is equal to _____.

$$\Rightarrow f_x(x, y)dx + f_y(x, y)dy$$

(35)

$P_0(x_0, y_0, z_0)$ be any point on the surface $f(x, y, z) = 0$, $f(x, y, z)$ is differentiable $P_0(x_0, y_0, z_0)$ then normal line at $P_0(x_0, y_0, z_0)$ has equation

$$\Rightarrow x = x_0 + f_x(x_0)t, y = y_0 + f_y(x_0)t, z = z_0 + f_z(x_0)t$$

(36)

A zero vector has magnitude equal to zero and _____ direction

→ No

(37)

Critical point which is neither Min nor Max

→ initial point

(38)

A continuous function on a set that is not closed and bounded set have any _____

→ absolute extrema

Q3 value $f(x_0, y_0)$ is relative $\max(f(x, y))$ domain D , exist and open $K \subset D$ radius is

$$\Rightarrow f(x_0, y_0) \geq f(x, y) \quad \forall x \in K$$

Q4 Two vector A and B are — if one vector A is scalar multiple of other vector B .

$\Rightarrow$ Parallel

Q5 $f_{xy} = f_{yx} = f_{yx}$ another notation

$$\Rightarrow \frac{\partial^3 f}{\partial x \partial y \partial x} = \frac{\partial^3 f}{\partial y \partial x \partial y} = \frac{\partial^3 f}{\partial x \partial y^2}$$

Q6 Cylindrical Co-ordinate system. point of space triple (r, θ, z) where (r, θ) represent

$\Rightarrow$ polar coordinate.

Q7 Partial derivative $f(x, y) = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j}$

$\Rightarrow$ grad

Q8 $f(x, y)$ the directional of $\vec{u}$ can be written

$$\nabla f(x, y) \cdot \hat{u}$$

Identify the variable
 $y = 2x + 1 \rightarrow$ Single variable
 $y = 5x + 6y + 10 \rightarrow$ Double variable
 $y = x^2 \rightarrow$ single

Short Questions Midterm function

(45) let $f(x, y) = x^2 + 9x^3y + 2y$
 point $(4, 2)$. if you, what is value, not

Ans $f(x, y) = x^2 + 9x^3y + 2y$
 $f(4, 2) = 16 + 9(4)^3(2) - 2(2)$
 $= 16 + 1152 - 4 = 1164$

(46) let $(x, y, z) = x^4 \cdot y \cdot e^{xyz}$ defined $(1, 3, 6)$

Ans $f(x, y, z) = x^4 \cdot y \cdot e^{xyz}$
 $= (1)(3)e^{(1)(3)(6)} = 1969799$

(47) $f(x, y, z) = \ln \sqrt{x^2 + y^2 + z^2}$ function at origin

Ans f is not defined at origin.

(48) at least two points that lie on parabola

Ans $y = x^2 - 6x + 8$ $x \geq 0$
 $x = 2, x = 4$
 $\begin{cases} x^2 - 4x - 2x + 8 \\ x(x-4) - 2(x-4) \\ (x-2)(x-4) \\ (2, 4) \end{cases}$

(49) Find mid point between $(6, 2, 4)$ $(2, 5, 8)$

Ans $(4, \frac{7}{2}, 6)$ $\Rightarrow (\frac{6+2}{2}, \frac{2+5}{2}, \frac{4+8}{2})$

(50) Find the distance between point A $(3, 4, 9)$ and B $(4, 5, 9)$

Ans (a) for line joining point
 $\sqrt{(4-3)^2 + (5-4)^2 + (9-9)^2} = \sqrt{17}$

Line joining Point $A(3, 2, 6)$, $B(0, 3, 4)$
 direction ratios of Line $AB = 0-3, 3-2, 4-6$
 $-3, 1, -2$

direction cosine of Line $AB = \frac{-3}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{-2}{\sqrt{14}}$

Q2 Find vertex point of Parabola $y = 2x^2 + 6x + 3$

Give equation of Parabola $= y = 2x^2 + 6x + 3$ $a=2$
 x -coordinate of vertex $= x_0 = \frac{-b}{2a} = \frac{-6}{2(2)} = \frac{-3}{2}$ $b=6$
 $c=3$

y -coordinate of vertex $= y_0 = 2(-\frac{3}{2})^2 + 6(-\frac{3}{2}) + 3$
 $= \frac{18}{4} - \frac{18}{2} + 3$
 $= \frac{-6}{4} = \frac{-3}{2}$ $(-\frac{3}{2}, -\frac{3}{2})$

Q3 Find Point where parabola $y = x^2 - 5x$ meet x -axis.

$y = x^2 - 5x$
 $x^2 - 5x = 0 \Rightarrow 0 = x(x-5) \Rightarrow x=0, x=5$
 Parabola meet x -axis at $(0, 0)$ $(5, 0)$

Q4 Find at least two points line on parabola $y = x^2 - 7x + 12$

$y = 0$, $x^2 - 7x + 12 = 0$
 $(x-4)(x-3) = 0$
 $x=4, x=3$

Two points of parabola $(4, 0)$ $(3, 0)$

⑤ $f(x,y) \geq 0$ on R then $\iint_R f(x,y) dA \geq 0$

$\Rightarrow f(x,y) \geq 0$

⑥ Double integration to find the volume of solid.

⑦ For each fixed y $c \leq y \leq d$ the function $f(x,y)$ is function of x alone then the area of function of interval $a \leq x \leq b$ is

$\Rightarrow A(y) = \int_a^b f(x,y) dx$

57 What is absolute maximum value of function below on interval $[-3, 0]$

$\Rightarrow f(x) = \sqrt{x+1}^2 + \frac{2x}{5}$

$\Rightarrow 0$

⑧ Reversing the order integration $\int_0^1 \int_{4x}^4 e^{-y^2} dy dx$

$\int_0^4 \int_0^{y/4} e^{-y^2} dx dy$

$\begin{cases} 4x \rightarrow y/4 \\ dy dx \Rightarrow dxdy \end{cases}$